

Study on Service Portfolio Optimization Considering Carbon Emissions in Cloud Manufacturing Environment

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Abstract—Cloud manufacturing is an advanced intelligent manufacturing model, and the cloud manufacturing service portfolio is a research hotspot in the cloud manufacturing system. In the past, the research on service portfolio model focused more on time, cost, quality, etc., and less on the carbon emissions generated by enterprises in the manufacturing process. Based on this, this paper establishes an optimization problem model with the constraints of minimum time, minimum cost, optimal service quality and minimum carbon emission level, and solves the model based on the NSGA-II algorithm, and finally proves the feasibility and applicability of the model and algorithm through a simulation case.

Keywords—cloud manufacturing, service collaboration, Carbon emissions, NSGA-II algorithm

I. INTRODUCTION

With the development of cloud computing, big data, virtual simulation and other technologies and their penetration in the manufacturing field, Li et al firstly put forward the concept of "cloud manufacturing"[1]. Cloud manufacturing is a new type of manufacturing model, based on the concept of "manufacturing as a service", combining emerging information technology, cloud computing, Internet of Things technology and existing manufacturing technology in the market to provide users with a highly shared service model of manufacturing resources.

With the development of relevant theories and the maturity of technology, many researchers have carried out in-depth research on the composition and scheduling of cloud services for multi-task manufacturing in the cloud manufacturing environment. Yuan et al. established a service quality index system for the manufacturing service portfolio[2]. Li et al. established a cloud service scheduling model from the perspective of tasks, and took time, cost and service quality as the optimization goals[3]. Yin et al. established a multi-objective optimization model with service time, service cost, service reliability, service quality and service matching as the optimization objectives to obtain the initial scheme of service combination[4]. In order to consider the impact of the social attributes of cloud manufacturing services on service collaboration, Wang et al. established a

measurement model of synergy between cloud services from the perspective of cooperation and competition[5]. Liu et al. studied a multi-objective optimization method of cloud manufacturing service portfolio interval based on service quality perception[6]. Some studies also consider the trust characteristics of cloud services to construct service composition or resource scheduling models, such as the reliability and trustworthiness of services[7] and the trust between cloud service enterprises[8].

Green and low-carbon is the key link to achieve industrial upgrading and high-quality economic development, and it is very important to enhance the green and low-carbon development capacity of industry in the context of "dual carbon". However, few researchers consider the carbon emissions generated by enterprises in the manufacturing process, so in order to consider the impact of carbon emission level in the cloud manufacturing service portfolio, this paper establishes a multi-objective optimization problem model including minimum time, minimum cost, optimal service quality and minimum carbon emission level, and solves the model based on the NSGA-II algorithm.

II. PROBLEM DESCRIPTION AND MODEL CONSTRUCTION

A. Problem Description

Platform managers, service demanders, and service providers are the three important roles in the cloud manufacturing model. The various manufacturing resources and manufacturing capabilities provided by the manufacturing resource providers will form a virtual manufacturing resource pool through virtualization and service service. When the service demander submits a manufacturing task on the cloud platform, the cloud platform will first decompose the manufacturing task of the service demander into executable subtasks, and search for matching services that meet the needs from the cloud manufacturing service platform according to the user's needs. Then, according to a certain logical order, the best services that meet the conditional constraints are selected and combined to complete the manufacturing task.

Suppose that in the production process of an enterprise in the cloud manufacturing platform environment, the

enterprise first publishes the manufacturing task on the cloud platform, and the cloud platform decomposes the tasks on the demand side to obtain n sub-tasks $Task = \{Task_1, Task_i, \dots, Task_n\}$, where i represents the number of sub-tasks. Then, the services that meet the functional requirements of each subtask are selected from the candidate service resources to form the subtask service set $S_{ij}, S_{ij} = \{S_{i1}, S_{i2}, S_{i3}, \dots, S_{im_i}\}$, and m_i represents the number of candidate services for the i th subtask. Finally, select a specific service from the set of candidate services corresponding to each subtask to execute the corresponding subtask.

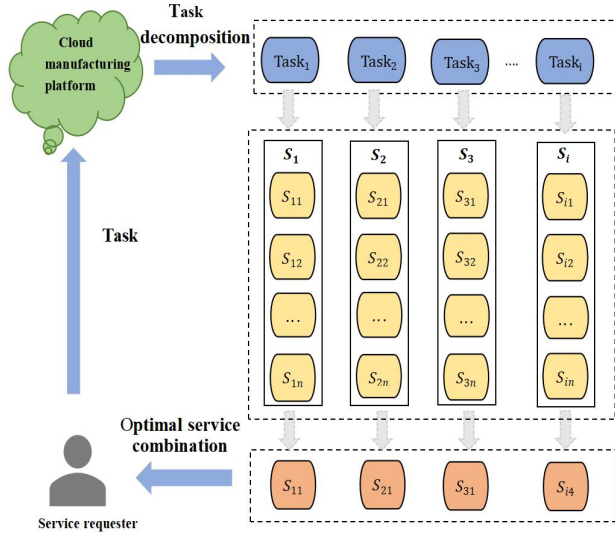


Fig. 1. Cloud manufacturing service portfolio optimization process

B. Model assumptions

Make the following assumptions regarding the issue in this article:

1) There are four different types of task structures in the cloud manufacturing environment, because the parallel structure, selection structure and circular structure can be converted into sequential structure through different rules, so this paper chooses to study the problem under the condition of sequential structure;

2) Each cloud service can process multiple subtasks, but only one subtask can be processed at the same time;

3) Each cloud service can be selected by multiple subtasks, but a subtask can only select one service to execute;

4) It is known in advance that the time, cost, and energy consumption of each subtask will vary from one cloud service to another.

C. Symbol definition

Table 1 shows the relevant symbol definitions of the model.

TABLE I. SYMBOL DEFINITION

symbol	meaning
S_{ij}	Candidate service for subtask
T_m	The manufacturing processing time for the subtask
T_s	The transit time from Candidate Service S_{ij} to S_{i+1j^*}
C_m	The cost of execution to complete the task
C_s	The cost of shipping from Candidate Service S_{ij} to S_{i+1j^*}
Q_{ij}	Service quality of Q_{ij}
CE_{ij}	The level of carbon emissions
x_{ij}	0-1 decision variable, if S_{ij} serves on the n th service, then it is 1; Otherwise, it is 0

D. Model building

Most of the existing service portfolio research models only consider QoS indicators, such as service time, service cost and service quality, and ignore the requirements of manufacturing tasks for the carbon emission level of enterprises. In fact, under today's "two-carbon" background, the carbon emission level has become an important indicator for the evaluation and selection of service competitiveness. Therefore, this paper introduces the carbon emission index into the model construction, and establishes an optimization problem model based on the constraints of minimum service time, minimum service cost, optimal service quality and minimum carbon emission level.

1) Service Hours

The time required to accept the demand side orders from the cloud manufacturing platform to deliver them to the demand side is mainly the execution time T_m and the transportation time T_s . Its objective function is:

$$\min T = \min(T_m + T_s) \quad (1)$$

$$T_m = \sum_{j=1}^{m_i} \sum_{i=1}^n T_m^{ij} x_{ij} \quad (2)$$

$$T_s = \sum_{j=1}^{m_i} \sum_{i=1}^n \sum_{j^*}^{m_{i+1}} T_s^{(ij, (i+1)j^*)} x_{ij} x_{(i+1)j^*} \quad (3)$$

2) Cost of Services

The total cost of each manufacturing service usually includes production costs, transportation costs, and cloud platform costs. Its objective function is:

$$\min C = \min(C_m + C_s) \quad (4)$$

$$C_m = \sum_{j=1}^{m_i} \sum_{i=1}^n C_m^{ij} x_{ij} \quad (5)$$

$$C_s = \sum_{j=1}^{m_i} \sum_{i=1}^n \sum_{j^*}^{m_{i+1}} C_s^{(ij, (i+1)j^*)} x_{ij} x_{(i+1)j^*} \quad (6)$$

3) Quality of service

The service quality of cloud services can be measured from the task qualification rate in entire manufacturing service process, and its objective function is:

$$\max Q = \max \left(\frac{1}{n} \sum_{i=1}^n \sum_{j=1}^{m_i} Q_{ij} x_{ij} \right) \quad (7)$$

4) Carbon emission levels

In the cloud manufacturing environment, it is very important to actively promote the green and low-carbon development of the manufacturing industry. In this paper, the carbon emission level is expressed as the mean value of the carbon emission level of each candidate service, and its objective function is:

$$\min CE = \min \left(\frac{1}{n} \sum_{j=1}^{m_i} \sum_{i=1}^n CE_{ij} x_{ij} \right) \quad (8)$$

Therefore, the total optimization objective function in this paper is:

$$F = (\min C, \min T, \min(-Q), \min CE) \quad (9)$$

$$\begin{cases} \min T = \min(T_m + T_s) \\ \min C = \min(C_m + C_s) \\ \min(-Q) = \min\left(-\frac{1}{n} \sum_{j=1}^{m_i} \sum_{i=1}^n Q_{ij} x_{ij}\right) \\ \min CE = \min\left(\frac{1}{n} \sum_{j=1}^{m_i} \sum_{i=1}^n CE_{ij} x_{ij}\right) \end{cases}$$

Based on the above metrics, the constraints that define the model are:

$$\text{s. t.} \begin{cases} T \leq T_{\max} \\ C \leq C_{\max} \\ Q \geq Q_{\min} \\ CE \leq CE_{\max} \\ \sum_{m=1}^M X_{ij}^m = 1, (i = 1, 2, 3 \dots I; j = 1, 2, 3 \dots J) \end{cases} \quad (10)$$

Among them, $T_{\max}$, $C_{\max}$, $Q_{\min}$, and $CE_{\max}$ respectively represent the maximum service time, maximum service cost, minimum service quality, and the highest carbon emission level of candidate service.

III. MODEL SOLVING

The model established in this paper is essentially a multi-objective combinatorial optimization problem, and there may be conflicting relationships between each optimization goal, which means that the improvement of some objective function values may weaken other objective functions, and all objective functions cannot be optimal, and only a set of Pareto solutions can be obtained. The NSGA-II algorithm is a commonly used multi-objective optimization algorithm, which is widely used to solve optimization problems with multiple conflicting objectives. It has the characteristics of few parameters, modularity and low complexity in the multi-objective optimization solution, so the NSGA-II algorithm proposed by Deb et al. is used to solve the model[9].

A. Algorithmic flow

The main steps of the NSGA-II algorithm are as follows:

Step 1: Initialize the parameters and set the evolutionary algebra $t=0$;

Step 2: The initial population P_t with the number of populations of N was randomly generated, and the rapid non-dominant ranking and crowding calculation were carried out on the population;

Step 3: The binary tournament method was used to select individuals from P_t and crossover and mutate to generate population Q_t ;

Step 4: Population P_t and population Q_t are merged into a new population R_t ;

Step 5: Perform fast non dominated sorting and crowding on R_t , and then select N individuals to form a new population P_{t+1} based on the elite retention strategy;

Step 6: Determine whether the algorithm meets the termination conditions, if not, iterate the third step to the fifth step repeatedly, otherwise the algorithm will run and output the Pareto solution set.

B. Algorithm design

1) encoding

Considering the characteristics of the research problem, this paper uses integer encoding to encode, as shown in Fig. 2. The meaning in the graph: the first subtask is executed by the third resource in the candidate service set S_{1j} , the second subtask is performed by the first resource in its candidate service set S_{2j} , and the third subtask is executed by the fourth resource in its candidate service set S_{3j} .

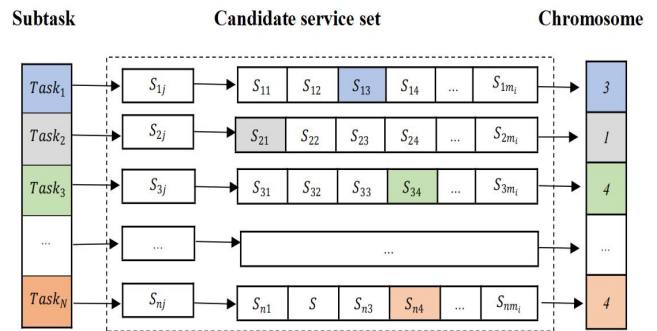


Fig. 2. Chromosome encoding method of NSGA-II algorithm

2) Selection operator

The selection operation mimics the law of "survival of the fittest" in nature, and if the individual's fitness is high, it has a greater probability of being passed on to the next generation, and vice versa. There are many ways to carry out the selection operation, and based on the model built above, this article uses a binary tournament selection strategy.

3) Crossover operator

The crossover operation of the NSGA-II algorithm simulates the cross-transposition of chromosomes in nature. In order to better inherit the excellent genes of the parents, this paper chooses to choose k-point crossing as the crossover operator.

4) Mutation operator

The variation operation introduces a certain randomness by changing the values of one or more genes in the solution to increase the search space. In this paper, we choose the reverse transformation method.

IV. EXPERIMENTAL SIMULATION AND ANALYSIS

A. Test example

Because it is difficult to obtain data from the cloud manufacturing platform in reality, this paper designs a simulation experiment based on extensive reading of relevant references. The cloud service platform encapsulates various types of manufacturing resources from multiple enterprises into virtual services and publishes them on the platform, and then decomposes the manufacturing tasks into 10 series sub-tasks according to the manufacturing requirements released by the service demander on the cloud platform. Due to the difference in machining accuracy requirements, each cloud service can process multiple types of sub-tasks, but only one sub-task can be processed at the same time. Table 2 lists the value ranges of the parameters.

TABLE II. THE RANGE OF VALUES OF RELEVANT PARAMETERS IN CLOUD MANUFACTURING

T_m/h	T_s/h	C_m/M	C_s/M	Q_{ij}	CE_{ij}
[71,119]	[8,30]	[0.15,0.65]	[0,0.5]	[0.6,1]	[0.5,1]

B. Solve and analyze examples

Based on the CPU configuration of AMD Ryzen 5 5500U with Radeon Graphics 2.10 GHz processor, the memory is 16GB on the computer, using Python programming for calculation. By referring to the relevant experimental results, the example parameters required in this paper are: the initial population size 100, the maximum number of iterations is set to 200, the crossover probability is set to 0.9, and variation probability is set to 0.1.

In the same running environment, the algorithm can obtain iterative results of four objective functions, as shown in Fig. 3. As can be seen from the Fig. 3, after several iterations, the fitness values of each target tend to stabilize and reach a convergence state. Then, the TOPSIS method is used to sort and screen the Pareto optimal service combination, and the best manufacturing scheme can be selected.

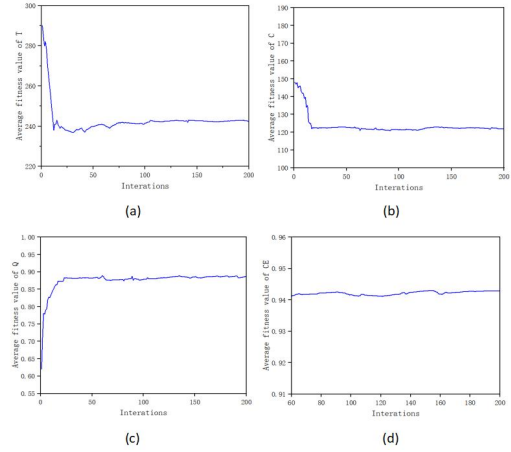


Fig. 3. The iterative result of the objective function

V. CONCLUSIONS

Most of the previous studies on cloud manufacturing service combination only consider the QoS attribute of cloud service, but this paper introduces the carbon emission level into the model building, establishing a multiple-target optimization model with minimum time, minimum cost, and optimal service quality and minimum carbon emission level. In order to achieve the optimization of the goals constructed in this paper at the same time, the NSGA-II algorithm is used to optimize the model. The effectiveness and feasibility of the proposed model and the solution algorithm in solving the cloud manufacturing service combination problem are proved.

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